

Basic Statistics

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1 Data Representation

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- Graphical
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- Central Tendency
- Measures of Dispersion
- Skewness
- Kurtosis

Chapter 3: Data Representation

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Data Representation

- Presentation of data refers to an exhibition or putting up data in an attractive and useful manner such that it can be easily interpreted.
- Three main forms of data presentation
 - Textual
 - Tabular
 - Graphical

Textual Representation I

Santoor has become the first soap brand from an Indian FMCG company to breach annual sales of Rs 2,000 crore. Wipro Consumer Care, the maker of Santoor, confirmed the number to TOI. With a turnover of over Rs 2,000 crore, Santoor has clearly overtaken HUL's soap brand Lux, and is now challenging the numero uno Lifebuoy. HUL's latest annual report places Lifebuoy and Lux in the Rs 2,000-crore and Rs 1,000-crore plus sales bracket, respectively.

According to industry sources quoting Kantar Household panel data, Santoor's all-India market share in January-March 2019, at 15.1%, has exceeded Lux's 12.5%, but is less than Lifebuoy's 17.7%. The urban market data, however, shows Santoor (13.4%) ahead of both Lux (12%) and Lifebuoy (13%). Kantar declined to comment on this data.

Industry sources quoting Nielsen data said Santoor (9.3%) is the third-largest brand after Lifebuoy (13.7%) and Lux (12%) for January-March 2019. Insights into data from Worldpanel Division of Kantar reveal that Santoor's penetration is much higher than Lux in South and parts of West regions. However, at a national level, Santoor has a much lesser penetration than Lux (34% against 60%).

Lux's penetration is driven by the Rs 10-pack (about 55g), with 60% of Lux-buying homes purchasing this pack. On the other hand, Santoor's penetration is driven largely by its 75g+ pack, with 70% of Santoor-buying homes purchasing this pack.

According to the data from Worldpanel Division of Kantar, Santoor also has a higher number of buying occasions than Lux (Santoor buyers purchase about 45% more times than Lux buyers). As a result, the overall volumes of Santoor have gone ahead of Lux in recent times.

Textual Representation II

- Most raw and vague form of presentation
- Used when the volume of data is small.

Tabular Representation I

- A table facilitates representation of even large amounts of data in an attractive, easy to read and organized manner.
- The data is organized in rows and columns.

Table 1: Soap sale data in Jan-March, 2019.

		Soap		
		Lifebuoy	Lux	Santoor
Company		Numero uno	HUL	Wipro
Sales volume in cr		2000	1000	2000
Market share percentage	Household, Kantar	17.1%	12.5%	15.1%
	Worldpanel, Kantar		60%	34%
	Urban market data	13%	12%	13.4%
	Nielsen data	13.7%	12%	9.3%
Most sold pack			60% 55g	70% 75g

- Components of Data Tables

- Table Number: should have a specific table number
- Title: tells the readers about the data it contains, time period of study, place of study, etc.
- Stubs: titles of the rows in a table
- Caption: title of a column in a table
- Body or field: content of a table in its entirety
- Headnotes and Footnotes: further aids in the purpose of a title
- Source: used for secondary data

- Construction of Data Tables

- The title should be in accordance with the objective of study
- If there might arise a need to compare any two rows or columns then these might be kept close to each other.
- If the rows in a data table are lengthy, then the stubs can be placed on the right-hand side of the table.
- Headings should be written in a singular form.
- A footnote should be given only if needed.
- Size of columns must be uniform and symmetrical.
- Headings and sub-headings should be free of abbreviations.
- There should be a clear specification of units above the columns.

Graphical Representation

- Data is displayed graphically
 - Easier for people to interpret the data
- Common plots are
 - Bar diagram, bar-chart, pie chart, histogram, etc.

Categorical Data Visualization

- Religions of 25 newborn babies:

Hindu, Muslim, Muslim, Buddhist, Christian, Sikh, Muslim, Buddhist, Muslim, Muslim, Buddhist, Buddhist, Christian, Hindu, Christian, Sikh, Christian, Christian, Christian, Buddhist, Buddhist, Christian, Hindu, Christian, Muslim

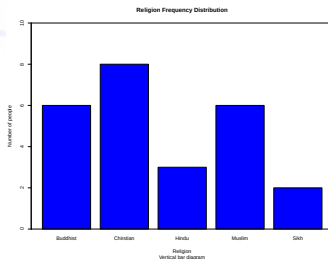
<i>Religion (Category)</i>	<i>Frequency</i>	<i>%</i>	<i>Angle(o)</i>
Buddhist	6	$6 \times (\frac{100}{25})\% = 24\%$	$6 \times (\frac{360}{25})^\circ = 86.4^\circ$
Christian	8	$8 \times 4\% = 32\%$	$8 \times 14.4^\circ = 115.2^\circ$
Hindu	3	$3 \times 4\% = 12\%$	$3 \times 14.4^\circ = 43.2^\circ$
Muslim	6	$6 \times 4\% = 24\%$	$6 \times 14.4^\circ = 86.4^\circ$
Sikh	2	$2 \times 4\% = 8\%$	$2 \times 14.4^\circ = 28.8^\circ$
<i>Total</i>	25	$25 \times 4\% = 100\%$	$25 \times 14.4^\circ = 360^\circ$

Basic Visualization II

- Vertical bar diagram

- It displays frequencies of categories of data.
- Categories in X-axis and frequencies in Y-axis
- Vertical bars of lengths proportional to the frequencies are drawn at each categories.

Vertical Bar Diagram



R-code

Basic Visualization III

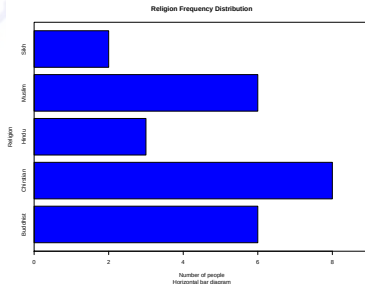
- 1 Create table from data:- `my_table = table(vector)`
- 2 Get diagram:- `barplot(my_table, xlab = ..., ylab = ..., ylim = ..., col = " ...")`
- 3 Add title:- `title(main = "", sub = "")`

Basic Visualization IV

- Horizontal bar diagram

- It displays frequencies of categories of data.
- Categories in Y-axis and frequencies in X-axis
- Horizontal bars of lengths proportional to the frequencies are drawn at each categories.

Horizontal Bar Diagram



Basic Visualization V

R-code

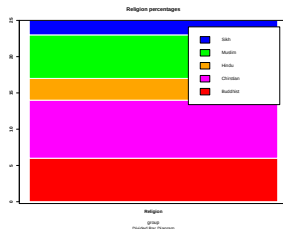
- 1 Create table from data:- `my_table = table(vector)`
- 2 Get diagram:- `barplot(my_table, xlab = ..., ylab = ..., ylim = ..., col = "...", horiz = T)`

Basic Visualization VI

- Divided bar diagram

- It shows the relative proportions of data in different categories within a bar.
- The size of the portion corresponding to the category A can be found as $\left(\frac{\text{\# data points in category } A}{\text{Total number of data points}} \times 100\% \right)$

Divided Bar Diagram



R-code

Basic Visualization VII

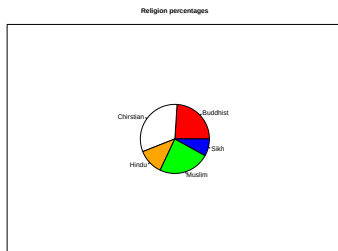
- 1 Create table from data:- `data = matrix(table(religion), nrow = 5); colnames(data) = c(Religion); rownames(data) = c(Buddhist, Chirstian, Hindu, Muslim, Sikh)`
- 2 Get diagram:- `barplot(data, col = c(red, magenta, orange, green, blue), xlab = . . . , legend.text = c(Buddhist, Chirstian, Hindu, Muslim, Sikh))`

Basic Visualization VIII

- Pie chart

- It shows the relative proportions of data in different categories within a circle.
- The size of an angle θ_A corresponding to the category A can be found as $\theta_A = \frac{\text{\# data points in category } A}{\text{Total number of data points}} \times 360^\circ$

Pie Chart



Basic Visualization IX

R-code

- 1 Create table from data:- `my_table = table(vector)`
- 2 `pie(my_table, col = . . . , clockwise = . . . , init.angle = 0)`
- 3 By default anti clockwise and initial angle is 0

Basic Visualization X

Discrete Data Visualization

- Number of siblings of 30 new born babies:

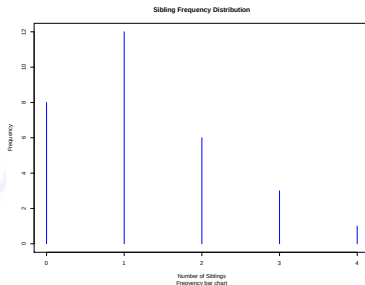
0, 2, 2, 1, 1, 1, 1, 3, 0, 0, 2, 1, 1, 0, 2, 3, 1, 4, 1, 2, 2, 0, 1, 0, 0, 1, 3, 0, 1, 1

Frequency Table

<i>Number of siblings</i>	<i>Frequency</i>	<i>Cumulative Frequency</i>
0	8	8
1	12	$8 + 12 = 20$
2	6	$8 + 12 + 6 = 26$
3	3	$8 + 12 + 6 + 3 = 29$
4	1	$8 + 12 + 6 + 3 + 1 = 30$
<i>Total</i>	30	-

- Frequency bar chart
 - It displays frequencies at different isolated values of data.
 - Isolated values in X-axis and frequencies in Y-axis
 - Perpendicular lines proportional to frequencies are drawn at isolated values.

Frequency bar chart

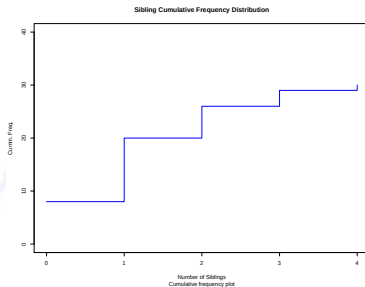


R-code

1 `plot(my_table, col = ..., xlab = ..., ylab = ...)`

- Cumulative frequency plot/ogive
 - Cumulative frequency on the y-axis and isolated values along the x-axis.

Cumulative frequency plot/ogive



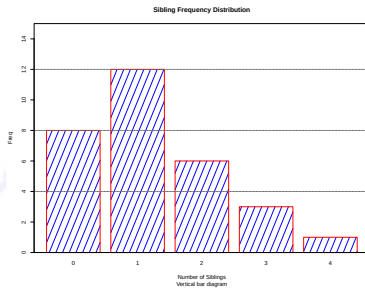
R-code

1 `plot(cumsum(my_table), col = ..., xlab = ..., ylab = ...)`

- Vertical bar diagram

- Isolated values in X-axis and frequencies in Y-axis
- Vertical bars of lengths proportional to the frequencies are drawn at each isolated value.

Vertical Bar Diagram



R-code

1 `barplot(my_table, xlab = ..., ylab = ..., ylim = ..., col = ..., border = ...)`

Continuous Data Visualization

- Weights of 60 newborn babies in Kg:

2.99, 2.74, 3.08, 3.04, 2.79, 2.63, 2.62, 3.40, 2.72, 2.53, 3.19, 2.77, 3.39, 3.67, 2.45, 2.41, 2.90, 3.50, 2.84, 3.55, 3.25, 2.56, 3.52, 3.03, 3.14, 3.07, 3.46, 3.13, 3.02, 3.15, 3.05, 3.20, 2.82, 2.89, 3.26, 3.01, 2.88, 3.01, 2.87, 2.70, 3.24, 3.74, 3.53, 3.34, 2.44, 3.72, 2.95, 3.09, 3.38, 3.16, 2.96, 2.39, 3.06, 2.86, 2.54, 2.94, 2.61, 2.48, 2.55, 2.62

- minimum weight = 2.39 kg
- maximum weight = 3.74 kg
- Range = $3.74 - 2.39 = 1.35$

Basic Visualization XVIII

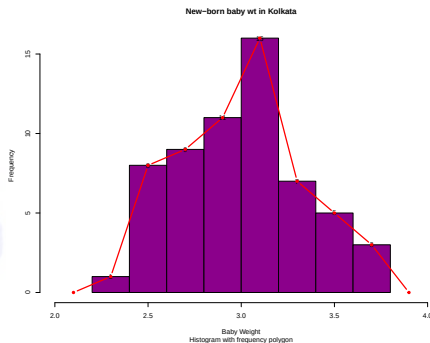
Frequency Table

<i>Class boundary</i>	<i>Class Mark</i>	<i>Freq. $\leq$ type</i>	<i>Relative Freq.</i>	<i>Area</i>	<i>Cum. Freq.</i>
2.0-2.4	2.2	1	$1/60/(2.4-2.0)$	1/60	1
2.4-2.8	2.6	17	$17/60/(2.8-2.4)$	17/60	18
2.8-3.2	3.0	27	$27/60/(3.2-2.8)$	27/60	45
3.2-3.6	3.4	12	$12/60/(3.6-3.2)$	12/60	57
3.6-4.0	3.8	3	$3/60/(4.0-3.6)$	3/60	60
<i>Total</i>	-	60		1	-

- Histogram

- It displays frequencies of quantitative data that has been sorted into intervals.
- Split the data into intervals, called bins
 - # of bins = $k \approx = (\max - \min)/h$, where h = bin width
 - $k \approx \sqrt{n}$
- Vertical bars of length proportional to the frequencies at each bins are drawn
- Sometimes relative frequency is given instead of absolute frequency

Histogram with Frequency Polygon



- Frequency polygon
 - Another type of frequency distribution graph
 - The number of observations is marked with a single point at the midpoint of an interval

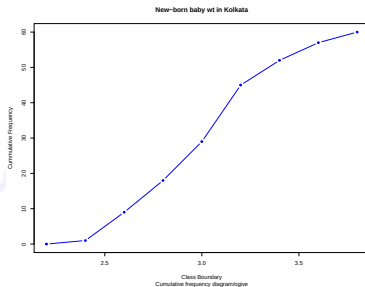
- A straight line then connects each set of points.

R-code

- 1 `hist(data, breaks = c(...), main = ..., xlab = ..., xlim = c(...), col = "", border = "", freq = FALSE)`
- 2 `lines(x = mid_values, y = freq, type = "", pch = "", col = "")`

- Cumulative frequency diagram/ogive
 - Cumulative frequency on the y-axis and class boundaries along the x-axis.

Cumulative frequency diagram/ogive

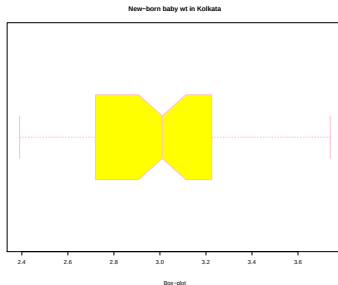


R-code

1 `plot(x = upper class limits, y = cumm. freq.)`

- Box plot (also known as a box and whiskers plot)
 - It displays the five 5 statistics $\max(\text{minimum}, \lceil Q_1 - 1.5 \times IQR \rceil)$, Q_1 , Q_2 , Q_3 , and $\min(\text{maximum}, \lfloor Q_3 + 1.5 \times IQR \rfloor)$, where $IQR(\text{Inter Quartile Range}) = Q_3 - Q_1$
 - $\lceil Q_1 - 1.5 \times IQR \rceil$: smallest data point higher than $Q_1 - 1.5 \times IQR$
 - $\lfloor Q_3 + 1.5 \times IQR \rfloor$: largest data point smaller than $Q_3 + 1.5 \times IQR$
 - The box can either be vertically or horizontally displayed depending on the labeling of the axis.

Box plot

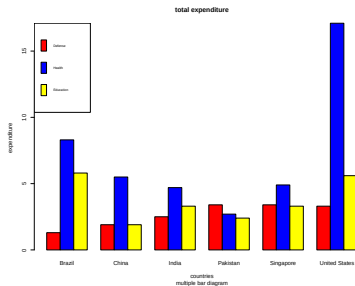
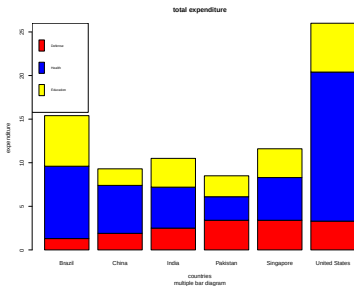


R-code

- 1 `boxplot(datat, col = "", border = "", horizontal =, notch =)`
- 2 By default it's vertical and without notch

- Multiple bar diagram
 - To compare different groups through charts

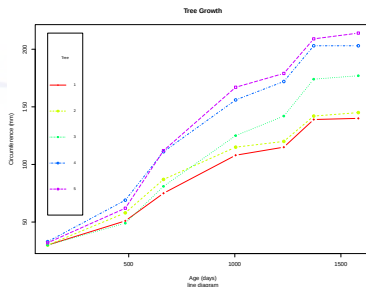
Multiple bar diagram



- Line diagram/ Line chart

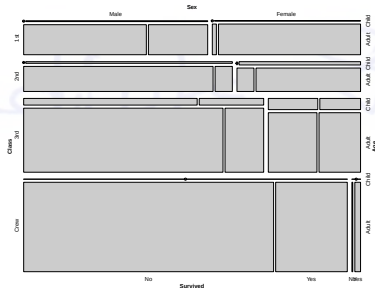
- A type of chart displaying information as a series of data points called 'markers' connected by straight line segments

Line diagram



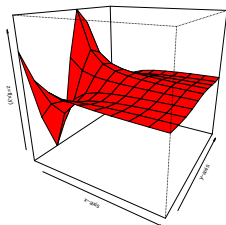
- Mosaic
 - To compare on more than one attribute

Mosaic



- 3-D plot

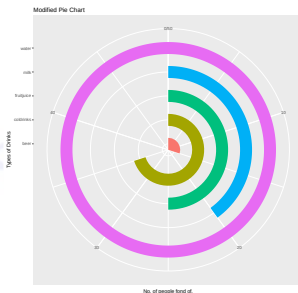
3-D Surface plot



- 3-D plot with control: Explore "*plotly*" package

- Modified pie-chart

Modified pie-chart



- Exploring "ggplot2" package

Chapter 4: Data Summary Measures

Sudipta Das

Summary Measures

- There are several types of summary measures which are useful for describing different aspects of the data (more specifically, different aspects of the distribution of data over the possible values).

Central Tendency I

- The first type is called measures of central tendency, which represent a central value for the collection of observations in the sample.
- Common measures of central tendency
 - 1 Mean
 - 1 Arithmetic
 - 2 Geometric
 - 3 Harmonic
 - 2 Median
 - Quartile, Decile, Percentile, Quantile
 - 3 Mode

(Arithmetic) Mean is the simple average of all the observations.

If a vehicle travels for a certain amount of time at a speed x and then the *same amount of time* at a speed y , then its average speed is the arithmetic mean of x and y

- For un-grouped or simple series data: n observations $(X_1, X_2, \dots, X_n)$
 - Mean is computed by dividing the sum of all the observations by the total number of observations.
 - So, the mean is $\frac{x_1+x_2+\dots+x_n}{n} (= \frac{1}{n} \sum_{i=1}^n x_i)$, denoted by $\bar{x}$.

Central Tendency IV

- For grouped data
 - Discrete variable:

Values(x)	x_1	x_2	$\dots$	x_i	$\dots$	x_k	Total
Freq(f)	f_1	f_2	$\dots$	f_i	$\dots$	f_k	N

- Continuous variable:

Class Interval	$x'_1 - x''_1$	$x'_2 - x''_2$		$x'_k - x''_k$	Total
Freq(f)	f_1	f_2	$\dots$	f_k	N
Class Marks(x)	x_1	x_2	$\dots$	x_k	-

where $x_i = \frac{x'_i + x''_i}{2}$.

- Mean is $\bar{x} = \frac{1}{N} \sum_{i=1}^k x_i f_i$

- Some properties of A.M.

- 1 Sum of deviations from AM is zero
- 2 If the variable X has mean $\bar{x}$ then the variable $Y = \frac{X-a}{b}$ has mean $\bar{y} = \frac{\bar{x}-a}{b}$
- 3 If two groups of observations $\{x_{1,1}, \dots, x_{1,n_1}\}$ and $\{x_{2,1}, \dots, x_{2,n_2}\}$ have AMs $\bar{x}_1$ and $\bar{x}_2$, respectively then
 - the combined mean of all the observations is

$$\bar{\bar{x}} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}.$$

- if $\bar{x}_1 < \bar{x}_2$,

$$\bar{x}_1 < \bar{\bar{x}} < \bar{x}_2$$

Geometric and Harmonic mean for *only* positive observations

- Geometric mean

$$\bar{x}_g = \begin{cases} \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}} & , \text{ungrouped} \\ \left(\prod_{i=1}^k x_i^{f_i} \right)^{\frac{1}{N}} & , \text{grouped,} \end{cases}$$

Average of probabilities

- Harmonic mean

$$\bar{x}_h = \begin{cases} \frac{\frac{n}{n}}{\sum_{i=1}^n \frac{1}{x_i}} & , \text{ungrouped} \\ \frac{\frac{N}{k}}{\sum_{i=1}^k \frac{f_i}{x_i}} & , \text{grouped,} \end{cases}$$

Average of speed: if a vehicle travels a certain distance d at a speed x and returns the *same distance* at a speed y , then its average speed is the harmonic mean of x and y .

- Some relations between A.M., G.M and H.M.
 - 1 The logarithm of the G.M. is equal to the A.M. of the logarithmic values.
 - 2 The reciprocal of the H.M is the A.M of the reciprocal values
 - 3 For any set of strictly positive real numbers, $A.M \geq G.M. \geq H.M.$
 - Equality hold when all numbers are equal

Median is the middle-most value among the ordered observations

- To compute median for series data
 - Arrange the observations $x_1, x_2, \dots, x_n$ from the smallest to the largest.
 - Then, if n is odd, the middle value (value of the $((n+1)/2)^{th}$ observational unit) is the median.
 - If n is even, the average of the two middlemost values (average of the values of $(n/2)^{th}$ and $(n/2 + 1)^{th}$ observational units) is the median.

Central Tendency X

- To compute median for grouped data:- discrete variable

Values(x)	x_1	x_2	$\dots$	x_i	$\dots$	x_k	Total
Freq(f)	f_1	f_2	$\dots$	f_i	$\dots$	f_k	N
Cum Freq($\leq$)	f'_1	f'_2	$\dots$	f'_i	$\dots$	f'_k	-

- Find the *first* class, whose cumulative frequency is greater than equal to $N/2$
 - If it is the i^{th} class then $\frac{N}{2} \leq f'_i$ and $f'_{i-1} < \frac{N}{2}$
- Then the median is $\tilde{x} = x_i$.

Central Tendency XI

- To compute median for grouped data:- continuous variable

Class Interval	$x'_1 - x''_1$	$x'_2 - x''_2$		$x'_i - x''_i$		$x'_k - x''_k$	Total
Freq(f)	f_1	f_2	...	f_i	...	f_k	N
Cum Freq($\leq$)	f'_1	f'_2	...	f'_i	...	f'_k	-

- Find the median class, i.e., *first* class, whose cumulative frequency is greater than equal to $N/2$
 - If it is the i^{th} class then $\frac{N}{2} \leq f'_i$ and $f'_{i-1} < \frac{N}{2}$
- Then the median is

$$\tilde{x} = x'_i + \frac{\frac{N}{2} - f'_{i-1}}{f_i} \times c.$$

A more general measure is Quantile

- The p^{th} quantile ($0 < p < 1$), of a frequency distribution Z_p is a value which divides the distribution in the ratio $p : (1 - p)$

- Different choices for p
 - $p = \frac{1}{4}, \frac{2}{4}, \frac{3}{4}$
 - $Z_{i/4} = Q_i$ is called i^{th} **quartile**
 - $Q_1 < Q_2 < Q_3$
 - $p = \frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \dots, \frac{9}{10}$
 - $Z_{i/10} = d_i$ is called i^{th} **decile**
 - $d_1 < d_2 < d_3 < \dots < d_9$
 - $p = \frac{1}{100}, \frac{2}{100}, \frac{3}{100}, \dots, \frac{99}{100}$
 - $Z_{i/100} = p_i$ is called i^{th} **percentile**
 - $p_1 < p_2 < p_3 < \dots < p_{99}$

- Common examples of quartiles, percentiles, deciles, etc..
 - In view of the brief discussion on percentiles before, the first quartile, denoted by Q_1 , is the 25th percentile.
 - Similarly, the second quartile, denoted by Q_2 , is the 50th percentile which is also the median.
 - The third quartile Q_3 is the 75th percentile.
 - The deciles are similarly defined as the 10th percentile, 20th percentile and so on.
 - In particular, the 5th decile is the 50th percentile, or the second quartile, or the median.

- To compute p^{th} quantile for series data
 - Consider the proportion and cumulative proportion for the ordered observations

Ordered Obs	$x_{(1)}$	$x_{(2)}$		$x_{(i)}$		$x_{(n)}$
Freq(f)	$1/n$	$1/n$	...	$1/n$	...	$1/n$
Cum Freq($\leq$) (proportion set)	$1/n$	$2/n$	...	i/n	...	1

- Look for the target proportion p in the proportion set.
 - if p is exactly one of the values in proportion set:
 $z_p = (\text{"that value"} + \text{"next value"})/2$
 - if p is crossed over the proportion set for the first time:
 $z_p = \text{"that value"}$

- To compute p^{th} quantile for grouped data
 - Find the class containing z_p , i.e., *first* class, whose cumulative frequency is greater than equal to Np
 - If it is the i^{th} class then $Np \leq f'_i$ and $f'_{i-1} < Np$
 - Then

$$z_p = x'_i + \frac{Np - f'_{i-1}}{f_i} \times c.$$

Mode is the value that occurs with highest frequency in the data

- Simple series: No mode
- Discrete grouped data

Values(x)	x_1	x_2	$\dots$	x_i	$\dots$	x_k	Total
Freq(f)	f_1	f_2	$\dots$	f_i	$\dots$	f_k	N

- If f_i is the unique maximum of $(f_1, f_2, \dots, f_k)$, then the mode is

$$\check{x} = x_i$$

- Note: There can be more than one mode.

- Continuous grouped data

Values(x)	$x'_1 - x''_1$	$x'_2 - x''_2$	...	$x'_i - x''_i$	...	$x'_k - x''_k$	Total
Freq(f)	f_1	f_2	...	f_i	...	f_k	N

- If f_i is the unique maximum of $(f_1, f_2, \dots, f_k)$, then $x'_i - x''_i$ is the modal class
- The mode is

$$\check{x} = x'_i + \frac{f_i - f_{i-1}}{2f_i - f_{i-1} - f_{i+1}} \times c$$

Measures of Dispersion I

- Measures of Dispersion describe the spread of the observations in the sample. More the spread, larger are these measures.
- Common measures of dispersion are range, interquartile range, variance, standard deviation, coefficient of variation, etc..

Dispersion Measures based on extreme values

1 Range

- Range is defined as the difference between the maximum value and the minimum value of the observations.
- For series data: $x_{(n)} - x_{(1)}$
- For grouped data:
 - Discrete: $x_k - x_1$
 - Continuous: $x_k'' - x_1'$

2 Interquartile range (IQR)

- $Q_3 - Q_1$, representing the range of values in which the middle 50% of the observations lie.
- Quartile deviation/Semi IQR: $(Q_3 - Q_1)/2$

Dispersion Measures based on all values

1 Mean deviation about A

- For series data: $MD_A = \frac{1}{n} \sum_{i=1}^n |x_i - A|$
- For grouped data: $MD_A = \frac{1}{N} \sum_{i=1}^k f_i |x_i - A|$
- Note:
 - If $A = \bar{x}$, it is called Mean deviation about mean and denoted by $MD_{\bar{x}}$
 - If $A = \tilde{x}$, it is called Mean deviation about median and denoted by $MD_{\tilde{x}}$

2 Root mean square deviation about A

- For series data: $RMSD_A = + \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - A)^2}$
- For grouped data: $RMSD_A = + \sqrt{\frac{1}{N} \sum_{i=1}^k f_i (x_i - A)^2}$
- Note:
 - If $A = \bar{x}$, it is called Standard deviation and denoted by $RMSD_{\bar{x}} = s$
 - The square of the standard deviation is called variance and denoted by s^2

- Simplified formula for variance

$$s^2 = \begin{cases} \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 & , \text{ungrouped} \\ \frac{1}{N} \sum_{i=1}^k f_i x_i^2 - \bar{x}^2 & , \text{grouped,} \end{cases}$$

Measures of Dispersion VI

- Some results on dispersion

- 1 Mean deviation about A is smallest when measured about median:

$$MD_{\bar{x}} \leq MD_A$$

Hint: $n MD_A = \sum_{i=1}^n |x_i - A| = |x_{(1)} - A| + |x_{(2)} - A| + \cdots + |x_{(n-1)} - A| + |x_{(n)} - A|$

- 2 Root Mean Square deviation about A is smallest when measured about mean:

$$RMSD_{\bar{x}} \leq RMSD_A$$

Hint: $n RMSD_A^2 = \sum_{i=1}^n (x_i - A)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - A)^2$

- 3 $s = 0 \Leftrightarrow$ all observations are equal

Measures of Dispersion VII

- ④ If the variable X having n values $(x_1, \dots, x_n)$ has s.d. s_x , then the variable $Y = a + bX$ has s.d. $|b|s_x$.
- ⑤ If two groups of observations $\{x_{1,1}, \dots, x_{1,n_1}\}$ and $\{x_{2,1}, \dots, x_{2,n_2}\}$ have AMs $\bar{x}_1$ and $\bar{x}_2$, respectively and have standard deviations s_1 and s_2 , respectively then the combined mean and sd of all the observations are

$$\bar{\bar{x}} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}.$$

$$s = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2} + \frac{n_1 (\bar{x}_1 - \bar{\bar{x}})^2 + n_2 (\bar{x}_2 - \bar{\bar{x}})^2}{n_1 + n_2}}$$

- ⑥ $MD_{\bar{x}} \leq RMSD_{\bar{x}}$

Hint: Take $a_i = |x_i - \bar{x}|$ and $b_i = 1 \forall i$ in the following CS inequality

$$(a_1 b_1 + \dots + a_n b_n)^2 \leq (a_1^2 + \dots + a_n^2)(b_1^2 + \dots + b_n^2)$$

Dispersion Measures based on mutual differences

- 1 Gini's Coefficients: a single number aimed at measuring the degree of inequality among values of a frequency distribution
- Gini's Coefficients = half of the relative mean absolute difference,

$$G = \frac{1}{2} \frac{\left(\frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n |x_i - x_j| \right)}{\bar{x}}$$

- Note:

- $G \geq 0$

- $G=0$ iff all observations have same values

- $G \leq 1$

- Hint : $|x_i - x_j| \leq |x_i| + |x_j|$

- $G \approx 1$ when all but one observations are zero

Measures of Dispersion X

Relative Measures of dispersion

1 Coefficient of variation (CV)

- It is SD/Mean (multiplied by 100 when expressed as a percentage)
- that is, $CV = \frac{s}{\bar{x}} \times 100$

2 Coefficient of mean-deviation (CMD)

- $CMD = \frac{MD_{\bar{x}}}{\bar{x}}$
- Can also be calculated w.r.t. $\tilde{x}$ and $\check{x}$

3 Coefficient of quartile deviation (CQD)

- $CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$

- Note: All of the above are unit-less

Skewness I

- Skewness is a measure of symmetry, or more precisely, the lack of symmetry.
- A frequency distribution, or data set, is symmetric if it looks the same to the left and right of the center point.
- For the n observations $x_1, x_2, \dots, x_n$, some measures of skewness are

① Fisher-Pearson coefficient of skewness:
$$\frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left(\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^{3/2}}$$

② Galton/Bowley's skewness: $(Q_3 - 2Q_2 + Q_1)/(Q_3 - Q_1)$

- Zero skewness: Symmetric
 - A frequency distribution, or data set, is symmetric if it looks the same to the left and right of the center point.
- Positive skewness: right skewed
 - Frequency distribution, or data set has longer right tail than left tail
- Negative skewness: left skewed
 - Frequency distribution, or data set has longer left tail than right tail

- Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution.
- For the n observations $x_1, x_2, \dots, x_n$, the formula for kurtosis is

$$\frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^2}.$$

- Mesokurtic:
 - For observations from normal data kurtosis is 3
- Leptokurtic:
 - a “positive” or thin distribution (fatter/heave tails), kurtosis is more than 3
 - Its does not imply the distribution is “tall” as sometimes reported.
 - It produces more outliers than the normal distribution.
- Platykurtic:
 - a “negative” or wide distribution (thin/lighter tails), kurtosis is lesser than 3
 - Its does not imply the distribution is “flat-topped” as sometimes reported.
 - It produces fewer and less extreme outliers than does the normal distribution